SUPPORTING ADULT GUIDANCE

# Tangents to a circle

Circle Theorems · Lesson 6. For any supporting adult — teacher, practitioner, parent or carer.
**You do not need to be a mathematics specialist to support this lesson.**

## What this lesson does

This lesson opens the tangent/chord family with two results: a tangent is perpendicular to the radius at the point of contact, and the two tangents from an external point are equal. Learners discover both by dragging the external point in the tangent-kite lab (right angles and equal lengths hold throughout), then prove the equal-tangents result by RHS congruence and reason the perpendicularity from the shortest-distance property. The kite relation ∠AOB + ∠APB = 180° is included for problem-solving.

Driving Question: *How does a line that only touches a circle behave?*

## The mathematics, in plain terms

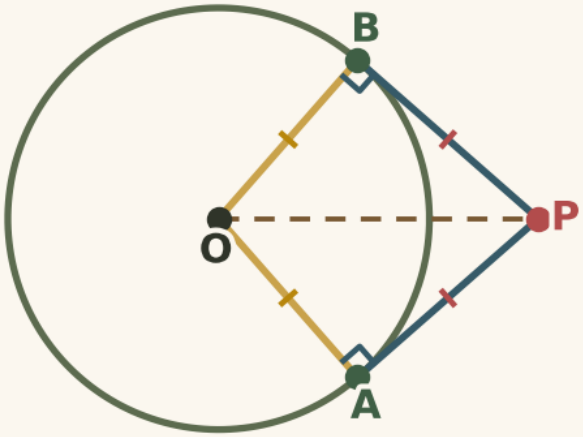
For a tangent touching at A with centre O, OA ■ tangent (the radius is the minimal distance from O to the line). For an external point P, the tangents PA and PB satisfy PA = PB: triangles OAP and OBP have OA = OB (radii), common hypotenuse OP, and right angles at A and B, hence are congruent (RHS). The quadrilateral OAPB is a right kite, so its angles give ∠AOB + ∠APB = 180°.

## The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

**Example 1 — Use the two tangent facts**

PA and PB are tangents from P, touching at A and B. The angle between them is ∠APB = 48°. Find ∠AOB, and state what you know about PA and PB.

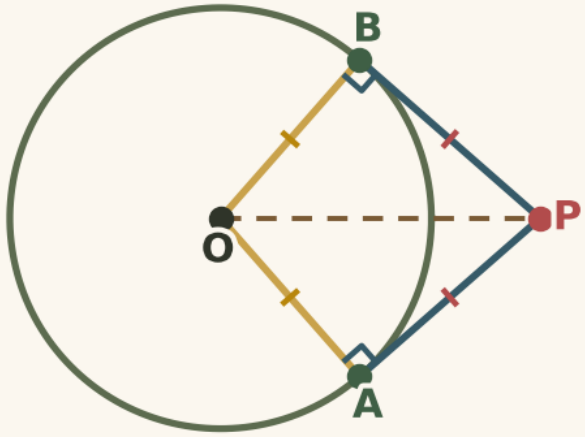



| | |
|---|---|
| **1.** Tangent meets radius at 90°: ∠OAP = ∠OBP = 90°. | right angles at A, B |
| **2.** Kite OAPB angles sum to 360°, so ∠AOB + ∠APB = 180°. | ∠AOB = 180 – 48 |
| **3.** The two tangents from a point are equal. | PA = PB |

**∠AOB = 132°, and PA = PB (the two tangents from P are equal).**

**Example 2 — Why the two tangents are equal**

Prove PA = PB for the two tangents from an external point P.




| | |
|---|---|
| **1.** Tangent ■ radius, so ∠OAP = ∠OBP = 90°. | right angles |
| **2.** OA = OB (radii) and OP is shared. | equal radii, shared side |
| **3.** Right angle, hypotenuse OP, one equal side ⇒ congruent (RHS). | triangles OAP ≅ OBP |

**So PA = PB — the two tangents from an external point are equal. ■**

## Two facts, both from equal radii

Let the learner drag the external point and see both invariants — the 90° angles and the equal lengths — before stating the rules. Prove the equal-tangents result cleanly with RHS congruence, and give the perpendicularity a reason (shortest distance) rather than asserting it. Insist on naming the fact used at each step, and introduce the kite angle relation once the two core facts are secure. This is the gateway to the alternate segment theorem (CT-07) and the perpendicular-from-centre-to-chord result (CT-08).

## The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

| Cornerstone | Its role here | Invitational timing |
|---|---|---|
| Connection | See the tangent kite: right angles and equal lengths. | ~9 min |
| Movement | Use the two facts (and the kite) to find angles and lengths. | ~11 min |

| | | |
|---|---|---|
| Reflection | Tangent-kite lab: drag P, both invariants hold. | ~12 min |
| Creativity | Prove PA = PB by RHS congruence; reason the 90°. | ~9 min |
| Rest | A short pause on symmetry as a shortcut. | ~2 min |
| Nutrition | Intellectual: geometry underwriting real motion and design. | ~3 min |

## Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

**Not spotting the right angle at the point of contact.**
'There is a radius to the point where the tangent touches — so there is a 90° angle there.'

**Forgetting the two tangents are equal.**
'Both tangents come from the same point P — so they are equal in length.'

**Giving a value with no reason.**
'Correct — and the reason is "tangent is perpendicular to the radius" or "the two tangents from a point are equal".'

**Assuming ∠AOB equals ∠APB.**
'In the kite they are supplementary, not equal: ∠AOB + ∠APB = 180°.'

## Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

**Please hold this line**
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

## Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: what a tangent is (Lesson 1), that all radii of a circle are equal, congruent triangles (RHS), and that angles in a triangle sum to 180°. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

## Supporting through questions: an example

Suppose a learner misses the right angle, or gives a length with no reason. The aim is a question that surfaces the radius, the equal tangents, and the reason.

- "Is there a radius drawn to the point of contact? Then what angle is hiding there?"
- "Both tangents come from the same point — what does that tell you about their lengths?"
- "What is your reason? Name the fact."

- “In the kite, how do ∠AOB and ∠APB relate — equal, or supplementary?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

## Questions you might have

**Why is the tangent perpendicular to the radius?**

Because the radius to the point of contact is the shortest line from the centre to the tangent, and the shortest line from a point to a line is always perpendicular to that line.

**Why are the two tangents equal?**

Triangles OAP and OBP have equal radii OA = OB, a shared hypotenuse OP, and right angles at A and B, so they are congruent (RHS). Their third sides PA and PB are therefore equal.

**What is the kite relation?**

OAPB is a kite with right angles at A and B. Since its four angles add to 360°, the remaining two satisfy ∠AOB + ∠APB = 180°.

**Where do tangents matter beyond the exam?**

A tangent is the instantaneous direction of travel around a curve — a wheel on a road, a belt on a pulley, an object flung from a spin all move along the tangent, perpendicular to the radius.

## Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

## Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

| Where they are | What it can look like |
|---|---|
| Approaching Proficiency | Beginning to engage; names some features and makes a start with support. |
| Developing Proficiency | Carries out the method with prompts; can explain part of the reasoning. |

| | |
|---|---|
| Proficiency | Works through independently and explains why each step is true. |
| Refining Proficiency | Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else. |

## A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

## If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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